

The top corners of the slide feature decorative geometric patterns. These consist of diagonal bands in shades of light purple and grey, with small white squares and rectangles interspersed within them.

# Geometry and Locus of Complex Numbers

Geometrical Interpretations in the Complex Plane

# Geometrical Interpretation in Complex Plane

In this section, we study the geometrical interpretation of a complex number  $Z$  in the complex plane and the locus of  $Z$  in Cartesian form.

## Example 2.18

Given the complex number  $z = 3 + 2i$ , represent the complex numbers  $z$ ,  $iz$ , and  $z + iz$  in one Argand plane. Show that these complex numbers form the vertices of an isosceles right triangle.

### Solution

Given that  $z = 3 + 2i$ .

Therefore,  $iz = i(3 + 2i) = -2 + 3i$

$z + iz = (3 + 2i) + (-2 + 3i) = 1 + 5i$

Let  $A$ ,  $B$ , and  $C$  be  $z$ ,  $z + iz$ , and  $iz$  respectively.

$$AB^2 = |(z + iz) - z|^2 = |-2 + 3i|^2 = 13$$

Since  $AB^2 + BC^2 = CA^2$  and  $AB = BC$ ,  $\triangle ABC$  is an isosceles right triangle.

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# Definition of a Circle in Complex Plane

## Definition 2.5 (circle)

A circle is defined as the locus of a point which moves in a plane such that its distance from a fixed point in that plane is always a constant. The fixed point is the centre and the constant distance is the radius of the circle.

## Equation of Complex Form of a Circle

The locus of  $z$  that satisfies the equation  $|z - z_0| = r$  where  $z_0$  is a fixed complex number and  $r$  is a fixed positive real number consists of all points  $z$  whose distance from  $z_0$  is  $r$ .

Therefore,  $|z - z_0| = r$  is the complex form of the equation of a circle.

- $|z - z_0| < r$  represents the points interior of the circle.
- $|z - z_0| > r$  represents the points exterior of the circle.

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# Circle Equation Examples

## Example 2.19

Show that  $|3z - 5 + i| = 4$  represents a circle, and, find its centre and radius.

### Solution

The given equation  $|3z - 5 + i| = 4$  can be written as:

$$3 \left| z - \frac{5-i}{3} \right| = 4 \Rightarrow \left| z - \frac{5}{3} + \frac{i}{3} \right| = \frac{4}{3}$$

It is of the form  $|z - z_0| = r$  and so it represents a circle, whose centre and radius are  $\frac{5}{3}$ ,  $-\frac{1}{3}$  and  $\frac{4}{3}$  respectively.

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# Cartesian Form of Locus

## Example 2.21

Obtain the Cartesian form of the locus of  $z$  in each of the following cases.

$$(i) \ |z| = |z - i| \quad (ii) \ |2z - 3 - i| = 3$$

### Solution (i)

$$\begin{aligned} \text{we have } |z| &= |z - i| \\ \Rightarrow |x + iy| &= |x + i(y - 1)| \\ \Rightarrow x^2 + y^2 &= x^2 + (y - 1)^2 \\ \Rightarrow x^2 + y^2 &= x^2 + y^2 - 2y + 1 \\ \Rightarrow 2y - 1 &= 0. \end{aligned}$$

# Cartesian Form of Locus (Cont.)

## Solution (ii)

Find locus for  $|2z - 3 - i| = 3$

### Step 1: Substitute z

we have  $|2z - 3 - i| = 3$

$$|2(x + iy) - 3 - i| = 3$$

Group real and imaginary parts:

$$|(2x - 3) + i(2y - 1)| = 3$$

### Step 2: Apply Modulus & Square

Squaring on both sides, we get

$$|(2x - 3) + i(2y - 1)|^2 = 9$$

$$\Rightarrow (2x - 3)^2 + (2y - 1)^2 = 9$$

**Resulting Equation:**

$$\Rightarrow 4x^2 + 4y^2 - 12x - 4y + 1 = 0$$

This is the locus of z in Cartesian form.

## Image Sources

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[https://matherama.com/Tutorials/Chapter%2004%20-%20Complex%20Numbers%20Basics/img/argand\\_plane.png](https://matherama.com/Tutorials/Chapter%2004%20-%20Complex%20Numbers%20Basics/img/argand_plane.png)

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